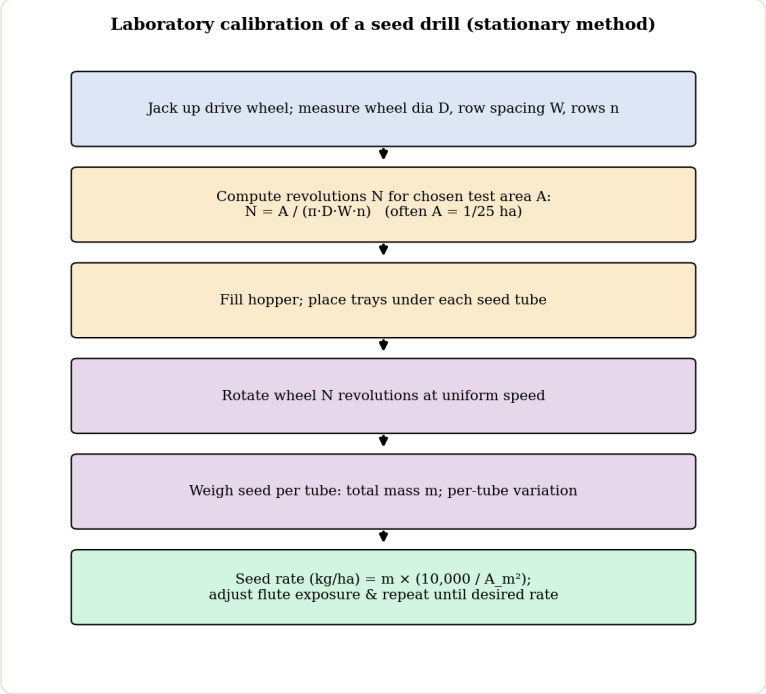
UNIT 2 · TOPIC 6(C)

Seed Drill Calibration

Syllabus topic: Design of soil working tools for sowing and planting

Calibration verifies that the drill delivers its design rate (a design-verification procedure).



Laboratory calibration of a seed drill (stationary method)

Procedure

1. Jack up the drive wheel; measure the wheel diameter D, row spacing W and number of rows n.
2. Compute the revolutions N for the chosen test area A (often A = 1/25 ha).
3. Fill the hopper and place a tray under each seed tube.
4. Rotate the wheel N revolutions at uniform speed.
5. Weigh the seed from each tube — total mass m and per-tube variation.
6. Calculate the seed rate; adjust the flute exposure and repeat until you get the desired rate.

$$N = A/(\pi \cdot D \cdot W \cdot n)$$

$$\text{Seed rate} = m/A;\ \text{in kg/ha} = m \times (10{,}000/A \text{ in } m^2)$$

- Correct for wheel slip (**5–10%**) when predicting field rates.
- Per-tube coefficient of variation: **CV ≤ 5–7%** is acceptable for drills.

Worked numerical. A drill with 9 rows at 20 cm spacing and a 1.2 m wheel, for 1/25 ha (400 m²): N = 400/(π × 1.2 × 0.20 × 9) = 400/6.79 ≈ **59 revolutions**.

Practice MCQs

Q143. Number of drive-wheel revolutions to simulate area A in drill calibration:

(a) A/(πDWn) (b) πDWn/A

(c) A·πDWn (d) πD/AWn

Q144. Acceptable coefficient of variation between seed-tube discharges of a drill is about:

(a) ≤5–7% (b) 15–20%

(c) 25–30% (d) 50%

Q145. A drill with 9 rows at 20 cm spacing, wheel dia 1.2 m: revolutions for 1/25 ha (400 m²) ≈

(a) 25 (b) 59

(c) 100 (d) 236

Answers

Q143 — (a) A/(πDWn)

Q144 — (a) ≤5–7%

Q145 — (b) 59 — *N = A/(π·D·W·n) = 400/(π × 1.2 × 0.20 × 9) = 400/6.79 ≈ 59 revolutions.*

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 11 (Crop Planting), Ch. 12 (Row-Crop Cultivation), Ch. 13 (Application of Fertilizers); standard FMP design references.